

Mohamed Abdelwahab

PhD Researcher in Robotics, Control, Planning, and Model-Based Reinforcement Learning

Padova, Italy | (+39) 3514725339 | mohamedabdelwahabb96@gmail.com |
linkedin.com/in/mohamed-abdelwahab-2605bb191

Research Profile

PhD researcher in Autonomous Systems working on **uncertainty-aware robot control, robot learning, and planning for robotic systems**. My research combines robust nonlinear control, Gaussian Process models, optimization-based planning/control, and model-based reinforcement learning for robotic manipulators and underwater vehicle-manipulator systems operating under uncertain dynamics.

Research Focus

- Model-based reinforcement learning for robotic control, with emphasis on Gaussian Process approximations, uncertainty propagation, and efficient policy optimization.
- Robust & adaptive control of manipulators under unknown dynamics, disturbances, and model mismatch.
- Hierarchical planning and control for underwater vehicle-manipulator systems.
- Sim2real transfer on benchmark systems and robotic platforms, including Franka Emika Panda and UVMS (BlueROV + Alpha5).

Education

University of Padova

PhD in Autonomous Systems (DAUSY)

Padova, Italy

Nov. 2023 – Present

- Research topics: model-based reinforcement learning for robotics; robust control for robotic manipulators; planning and control of UVMS in confined environments.

University of Padova

MSc in Control Systems Engineering, final grade: 110/110

Padova, Italy

Sept. 2021 – Sept. 2023

- Thesis: *Robotic Manipulator Control in the Presence of Uncertainty*.

Mansoura University

BSc in Mechatronics Engineering, final grade: 3.34/4, Very Good with Honors

Mansoura, Egypt

Sept. 2013 – June 2018

- Thesis: *Multi-Task Industrial Robot Arm*.

Research and Teaching Experience

Norwegian University of Science and Technology (NTNU)

Visiting Researcher

Trondheim, Norway

Sept. 2025 – Present

- Develop simulation environments for surface and underwater marine manipulation planning & control tasks.
- Apply model-free/model-based reinforcement learning and build evaluation pipelines for robustness, sample efficiency, and sim-to-real transfer.

University of Padova

Tutor, Adversarial Machine Learning

Padova, Italy

Oct. 2023 – Jan. 2025

- Supported course delivery, student supervision, and technical explanation of machine-learning robustness topics.

Selected Research Projects

P-MC-PILCO: Computationally Efficient Model-Based RL via GP Approximations

Model-based reinforcement learning; Gaussian Processes; robot control

- Developed a GP-based model-based RL framework that reduces stochastic rollout cost using approximate GP models and weight-space sampling.
- Validated on cart-pole, double pendulum on cart, and swimmer benchmarks; deployed on a real 7-DoF Franka Emika Panda for high-frequency trajectory tracking.

Hierarchical Topology-Aware Planning and Control of UVMS in Confined Environments

Underwater vehicle-manipulator systems; topology-aware planning; Pinocchio; Gazebo

- Developed a three-layer planning-and-control architecture for confined underwater intervention.
- Designed corridor-conditioned manipulation-aware refinement for joint base/arm optimization and P-MC-PILCO for tracking and station keeping.

Robust Control Using Gaussian Process Regression for Robotic Manipulators

Robust control; Gaussian Process Regression; manipulator dynamics

- Learned residual torque errors with GPR after nominal manipulator dynamics identification.
- Used GP predictive uncertainty as an online estimate of modeling error and as a basis for robust controller design.

Robust Controllers for Different Robotic Manipulators

Feedback linearization; sliding mode control; nonlinear systems

- Extended robust feedback linearization and sliding mode control to RR planar, SCARA, and UR10 models, comparing tracking, control, and robustness under model uncertainty.

Publications and Manuscripts

1. **Mohamed Abdelwahab**, Damiano Varagnolo, Ruggero Carli, Alberto Dalla Libera. *Hierarchical Topology-Aware Planning and Control of Underwater Vehicle–Manipulator Systems in Confined Environments*. Submitted to Ocean Engineering Journal.
2. **Mohamed Abdelwahab**, Niccolo Turcato, Giulio Giacomuzzo, Ruggero Carli, Alberto Dalla Libera. *Computationally Efficient Model-Based Reinforcement Learning for Robot Control via Gaussian Process Approximations*. Submitted to Journal of Intelligent & Robotic Systems.
3. Giulio Giacomuzzo, **Mohamed Abdelwahab**, Alberto Dalla Libera, Ruggero Carli. *A Robust Controller Based on Gaussian Processes for Robotic Manipulators with Unknown Uncertainty*. In *2025 33rd Mediterranean Conference on Control and Automation (MED)*, 2025.
4. **Mohamed Abdelwahab**, Giulio Giacomuzzo, Alberto Dalla Libera, Ruggero Carli. *Adaptive Robust Controller for Handling Unknown Uncertainty of Robotic Manipulators*. In *2024 IEEE 20th International Conference on Automation Science and Engineering (CASE)*, 2024.

Awards and Competitions

Bosch Future Mobility Challenge 2023

2nd Place, Team DEI UNIPD

International
2023

- Developed autonomous-driving algorithms for a 1/10-scale smart-city vehicle, including lane keeping, intersections, roundabouts, obstacle/pedestrian avoidance, overtaking, and parking.

Technical Skills

Robotics & Control	Manipulation; planning; robust & adaptive control.
Robot Learning	Model-based & model-free RL; Gaussian Process dynamics modeling; uncertainty propagation; sample-efficient learning; sim-to-real transfer.
Planning & Optimization	Sampling-based planning; system identification; uncertainty estimation.
Software & Tools	Python; MATLAB; C/C++; ROS2; Linux; Git; PyTorch; Pinocchio; Gazebo; Simulink.
Languages	Arabic; English.

References

Prof. Alberto Dalla Libera, Email: alberto.dallalibera@unipd.it

Prof. Ruggero Carli, Email: ruggero.carli@unipd.it

Prof. Damiano Varagnolo, Email: damiano.varagnolo@ntnu.no